

FDE Xlerate

Week 1, Day 1

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The problem

■ Task

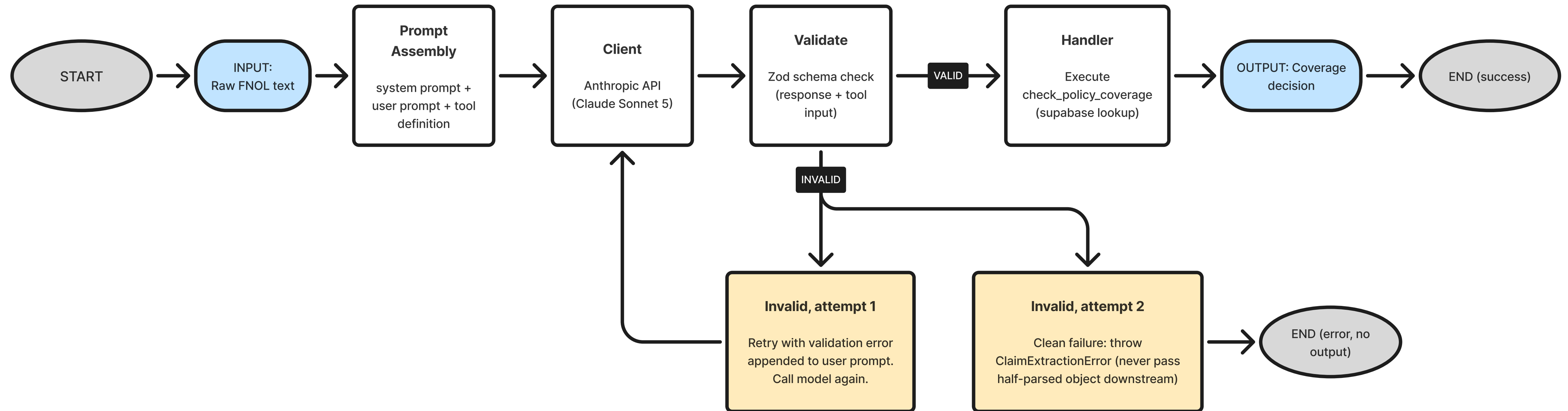
- Build the smallest honest version of an LLM-backed feature: structured output with exactly one function call
- No loop, no agent - feel where the model is and isn't reliable

■ Business problem

- Industry: insurance
- Case: First-notice-of-loss peril identification, extraction, and comparison against policy coverage

When an insurance customer submits a claim, the submission includes a plain text description of the incident. From this plain text, a categorical 'peril' must be identified and confirmed as covered or not covered by the customer's insurance plan.

High-level flow



Architecture

Client Layer

Owns the model API call and config.

Schema Layer

Single source of truth for “valid”:

- The Zod response contract (peril, loss_date, estimated severity, coverage_flag)
- The ‘check_policy_coverage’ tool definition

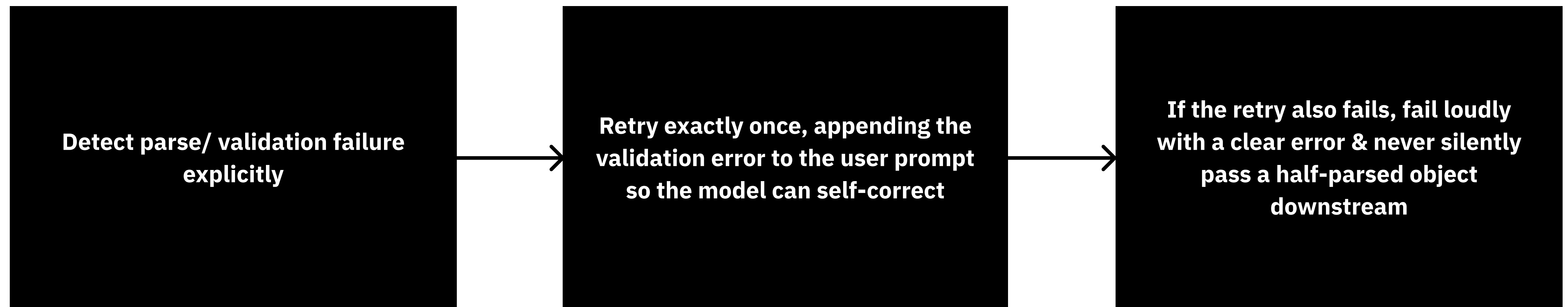
Orchestrator

- Assembles the three inspectable pieces:
 - System prompt
 - User prompt
 - Tool definition
- Owns the one-retry loop
- Logs raw & parsed response on every attempt

Handler Layer

- Executes the function the model asks for
- Never talks to the model
- Note: policy number is sourced separately (auth session, CRM, separate field) rather than extracted.

Malformed-Output Handling



■ Forcing a malformed output

Adversarial prompting could not reliably trigger a malformed output due. I changed the model from Claude Sonnet 5 to Claude Haiku 4.5. Haiku 4.5 consistently wrapped valid JSON in markdown code fences and failed differently on retry.

Tech Stack & key decisions

■ Claude Sonnet 5

- Bounded extraction-plus-one-tool-call task
- Opus' extra cost buys nothing
- Haiku repeatedly failed in testing by wrapping JSON in markdown fences, even after retry with error message appended to prompt.
- A client may not allow use of this model due to
 - Existing vendor relationships with other model providers
 - Lack of regulatory/ compliance approval for this model with existing approval for another model
 - Data residency and model hosting location
 - Mismatch with internal platform standardization

■ Supabase

Policy lookup via service-role key (server-side only, bypasses RLS by design)

■ TypeScript + Zod

Response contract and tool input/ output validated in-app

■ Pino

Structured logging so raw/ parsed responses are queryable after the fact

■ Max_tokens

Set to 1024 (~5x output token volume in testing)

■ Temperature

Set to 1 (temperature deprecated for Claude models after Claude Opus 4.6); otherwise, set to 0.